

AetherVault: Builder Grant Proposal

Protocol for Decentralized Digital Legacy & Intellectual Property

1. Executive Summary

AetherVault is an end-to-end encrypted protocol suite built on the BNB Chain, offering a comprehensive ecosystem for securing digital assets. Our solution integrates **Aether Capsule** (time-locked encryption), **Legacy Vault** (automated dead-man's switch), and **Aether Proof™** (on-chain intellectual property registry) to provide users with a secure, permanent digital legacy.

2. Technical Infrastructure & Tech Stack

Our robust architecture leverages a industry-leading stack to ensure both security and scalability:

- **BNB Chain:** Serves as the core logic layer, handling smart contract execution, ownership verification, and high-frequency state management.
- **Irys / Arweave:** Provides perpetual, decentralized storage to ensure encrypted file payloads and asset metadata survive forever, independent of central servers.
- **ECIES / Secp256k1:** Delivers client-side, zero-knowledge encryption, ensuring that no plaintext data ever leaves the user's browser.

3. Tokenomics, Real-Yield Staking & Trustless Burn

AetherVault operates on a deflationary circular economy powered by the \$AETH token:

- **Trustless On-Chain Burn:** Protocol interaction fees (minting, certification, storage access) trigger automated, immutable token burns directly via smart contracts, ensuring absolute transparency with zero central manipulation.

- **Real-Yield Staking:** Staking rewards are derived from protocol revenue (Real Yield). Rewards are non-inflationary, funded entirely by actual platform usage, which protects the token ecosystem from inflationary collapse and ensures long-term sustainability.

4. Strategic Alignment with BNB Chain

By building natively on the BNB Chain, AetherVault benefits from superior transaction speeds and industry-leading low gas costs. This allows our protocol to offer high-frequency, complex interactions—such as automated "Proof of Life" pings and time-locked vault management—at scale. We are committed to bringing high-utility, IP-focused dApp innovation to the BNB Chain user base.